

Module 3: Probability Basics

Random Variables

A random variable is a function that assigns a numerical value to each outcome of a random experiment. Discrete random variables take countable values, such as the result of rolling a die, while continuous random variables can take any value within an interval, such as a person's height. The behavior of a random variable is fully described by its probability distribution, which specifies how likely each possible value or range of values is.

Expected Value

The expected value of a random variable is the long-run average outcome if an experiment were repeated infinitely many times. For a discrete random variable X with possible values x_i and probabilities p_i , the expected value is computed as the sum of x_i times p_i over all i . The expected value is a measure of central tendency and forms the basis for more advanced concepts such as variance and covariance.

Variance and Standard Deviation

Variance measures how far a set of values is spread out from their expected value. It is defined as the expected value of the squared deviation from the mean. The standard deviation is the square root of the variance and is expressed in the same units as the original random variable, which makes it easier to interpret than the variance itself when describing the spread of a distribution.